

## **A Critical Review of CRISP-DM and the Laws of Data Mining**

This paper breaks down the two things that are crucial for data mining: the CRISP-DM process and the Nine Laws of Data Mining (McCormick, 2024).

### **The Six Steps of CRISP-DM**

CRISP-DM stands for Cross-Industry Standard Process for Data Mining. It was developed in the late 1990s and is a proven way to structure data mining or data science work. I used this methodology all the time to keep my projects on track. It has six phases. They are:

#### **Business Understanding**

In this step, we must define what the business needs. This could be writing down requirements in a requirements specification document.

#### **Data Understanding**

Collect data and explore it to see what it contains and get a quick view of data quality.

#### **Data Preparation**

In this step, we clean, transform, and shape the data for analysis.

#### **Modeling**

Here, we build models on the analyzed data. This is where actual data mining or machine learning algorithms get implemented.

#### **Evaluation**

This is a critical step to check the quality of the model and to evaluate the model with metrics such as “Does the model actually solve the business problem?”

#### **Deployment**

This is the final step to put the model for wider public use and to make sure it works as expected through continuous monitoring and support.

## The Nine Laws of Data Mining

Data mining expert Tom Khabaza created nine laws, which are simple truths explaining why the process works the way it does (McCormick, 2024):

1. Business goals drive everything.
2. We need business knowledge at every step.
3. Data preparation takes the most time.
4. Patterns always exist in the data.
5. Find the best model through experimentation.
6. Data mining helps you see better.
7. Prediction adds new information.
8. Value comes from business impact, not just accuracy.
9. As data and business goals keep evolving, so does the model.

### Quick Comparison: CRISP-DM, SEMMA, and KDD

CRISP-DM, SEMMA (Sample, Explore, Modify, Model, Assess), and KDD (Knowledge Discovery in Databases) are three standard data mining methods.

CRISP-DM is a widely used method and stands out for its clear steps and focus on business needs. It's flexible, well-documented, and fits many industries. One catch: the original version didn't say much about rolling out and watching models, but updates now fill that gap. I once used CRISP-DM for a retail project; it helped me keep business goals front and center.

SEMMA, from SAS, zooms in on technical details. It's easy to use with SAS tools but it skips over business context. So, it's not ideal for projects where business goals take priority (SAS Institute, 2025).

KDD covers everything from picking data to making decisions out on results. It's broad but sometimes feels too abstract for day-to-day use (Fayyad et al., 1996).

### Recent Innovations in CRISP-DM

CRISP-DM keeps evolving. The CRISP-DM 2.0 initiative brings in big data, real-time analytics, and artificial intelligence. Plus, agile project management is making the process more flexible. Microsoft's Team Data Science Process (TDSP) and IBM's Analytics Solutions Unified Method (ASUM-DM) build on CRISP-DM for their cloud and enterprise needs (Martínez-Plumed et al., 2019).

### Ethics & Biases in CRISP-DM

Every step in data mining can trip us up with bias or ethical issues. We need smart strategies to mitigate them as outlined in the table below:

**Table 1**

*Ethical Issues and Mitigation Strategies Across the CRISP-DM Phases*

| CRISP-DM Phase         | Ethical Issues & Potential Biases                                | Strategies for Ethical Practice                                                                    |
|------------------------|------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| Business Understanding | Goals that might exclude people (e.g., discriminatory outcomes)  | Set up an ethics board. Focus on fairness. (e.g., "find creditworthy people in neglected groups"). |
| Data Understanding     | Past data with hidden bias                                       | Audit for bias. Use fairness tools.                                                                |
| Data Preparation       | Features that amplify bias (e.g., ZIP code as a proxy for race). | Document changes. Reweigh data if needed.                                                          |
| Modeling               | Algorithms that worsen bias                                      | Test simple models. Add fairness checks.                                                           |
| Evaluation             | Metrics that hide unfair results                                 | Use fairness metrics. Check results by group (e.g., gender, race).                                 |
| Deployment             | Black box models that lack trust                                 | Develop explainable models (XAI). Add human review. Monitor often.                                 |

## Conclusion

CRISP-DM and the Nine Laws give us a solid foundation to lead any data mining or data science project with confidence. The process outlined in this document keeps the work tied to business goals and states the ethical guiding principles for all major issues. As data science grows with Agile and MLOps, CRISP-DM's focus on business goals and iteration ensures project scope stays relevant and aligned. The guidelines and the strategies discussed in this paper emphasize that the solutions we build are not just smart, but also responsible and fair (McCormick, 2024).

## References

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